ANYING GRACE

Business Computing

Regno.2025/DBC/DAY/0905/G

Assignment

QUESTION 1

```
Public Class frmLogin
    Private Sub btnLogin_Click(sender As Object, e As EventArgs) Handles btnLogin.Click
        Dim username As String = txtUsername.Text
        Dim password As String = txtPassword.Text

        'Simple validation
        If username = "student" And password = "5678" Then
            MessageBox.Show("Login Successful!", "Success")
            Form2.Show()
            Me.Hide()
        Else
            MessageBox.Show("Invaild username or password!", "Error")
            txtusername.Clear()
            txtpassword.Clear()
            txtusername.Focus()

        End If
    End Sub
End Class
```

QUESTION 2: Shopping Mall Discount

```
Sub Main()
    Dim totalAmount As Double
    Dim discountsRate As Double
    Dim finalAmount As Double

    'Loop to process 3 customers
    For i As Integer = 1 To 3
        Console.Write("Enter total amount for Customer " & i & ": $")
        totalAmount = Convert.ToDouble(Console.ReadLine())

        'If...ElseIf...for discount calculation
        If totalAmount < 100 Then
            discountsRate = 0
        ElseIf totalAmount >= 100 And totalAmount <= 500 Then
            discountsRate = totalAmount * 0.1 '10% discount
        Else
            discountsRate = totalAmount * 0.2 '20% discount
        End If

        finalAmount = totalAmount - discountsRate

        'Display results
        Console.WriteLine("----- Customer " & i & " Receipt ------")
        Console.WriteLine("Total Amount: $" & totalAmount.ToString("F2"))
        Console.WriteLine("Discount Applied: $" & discountsRate.ToString("F2"))
        Console.WriteLine("Final Amount to Pay: $" & finalAmount.ToString("F2"))
        Console.WriteLine()

    Next

    Console.ReadLine()
End Sub
```

QUESTION 3: OOP in VB

a) Class and Object

A Class is a blueprint or template that defines properties (data) and methods (behavior) of an object. An Object is an instance of a class.

b) Inheritance vs Polymorphism

- Inheritance: A mechanism where a child class inherits properties and methods from a parent class. It promotes code reuse. E.g., Car inherits from Vehicle class.

- Polymorphism: Means "many forms" - same method name behaves differently in different classes. E.g., Parent class has Drawn () method, Child classes Circle and Square both override Draw () to draw differently.

c) Encapsulation

Encapsulation is bundling data and methods that operate on that data into a single unit (class) and hiding internal details from outside world using access modifiers (Public, Private, Protected).

Importance

1. Data security

2. Prevents unauthorized access

3. Easy maintenance

4. Controls how data is modified.

d) Constructors and Destructors Constructor (Sub New): A special method that runs automatically when an object is created. Used to initialize properties.

Destructor (Sub Finalize): Runs when object is destroyed. Used to clean up resources like closing files or database connections.

QUESTION 4: VB Concepts

a) Three Uses of Collections in VB

1. Storing multiple items: Store group of similar objects like list of student names without creating many variables. E.g., Array List, List.

2. Dynamic data management: Add, remove, search items easily at runtime. E.g., shopping cart items in a mall app.

3. Iterating and Sorting: Loop through items, sort, count. E.g., displaying all customers in a List Box.

b) Message Boxes and their Types

Message Box is used to display messages and get user response. Types:

-Message Box Style. Information: Shows info icon.

- Message Box Style. Exclamation / Warning: Shows warning icon.

-Message Box Style. Critical: Shows error icon.

- Message Box Style. Yes, No, Cancel / Cancel: Shows buttons for user choice.

- Example: Message Box. Show ("Invalid marks!", "Error", MessageBoxButtons. OK, MessageBoxIcon. Error)

c) **Solution Explorer**

Solution Explorer is a window that shows the structure of your project. It helps in:

1. Managing files: View all forms, modules, and classes.

2. Adding/removing items: Add new forms, references.

3. Organizing project: Create folders, rename files.

4. Quick navigation: Double-click to open code.

QUESTION 5: Student Grade Management System

Part A: Input and Validation

```vb
Public Class StudentInfo
    ' This program collects student details and displays them.
    Private Sub btnShow_Click(sender As Object, e As EventArgs) Handles btnShow.Click
        'Declare variables
        Dim name As String = txtName.Text.Trim()
        Dim age As Integer
        Dim course As String = txtCourse.Text.Trim()
        ' Validate and convert age
        If Integer.TryParse(txtAge.Text.Trim(), age) Then
            'Display student details
            Dim message As String = "Name: " & name & vbCrLf & _
                                    "Age: "& age&vbCrLf & _
                                    "Course: " & course
            MessageBox.Show(message, "Student Information", MessageBoxButtons.OK, MessageBoxIcon.Information)
        Else
            MessageBox.Show("Please enter a valid age!", "Input Error", MessageBoxButtons.OK, MessageBoxIcon.Error)
        End If
        ' Clear the text boxes
        txtName.Clear()
        txtAge.Clear()
        txtCourse.Clear()
    End Sub
End Class
```

Part B: Calculations and Grading (8 marks)

```
rivate Sub CalculateGrades()
    Dim ict As Double = Val(txtICT.Text)
    Dim vb As Double = Val(txtVB.Text)
    Dim res As Double = Val(txtResearch.Text)
    Dim cyber As Double = Val(txtCyberLaw.Text)

    Dim total As Double = ict + vb + res + cyber
    Dim average As Double = total / 4
    Dim grade As String
    Dim status As String

    ' Determine letter grade
    If average >= 90 Then
        grade = "A"
    ElseIf average >= 80 Then
        grade = "B"
    ElseIf average >= 70 Then
        grade = "C"
    ElseIf average >= 60 Then
        grade = "D"
    Else
        grade = "F"
    End If

    ' Pass/Fail status
    If average >= 60 Then
        status = "PASS"
    Else
        status = "FAIL"
    End If
```

Part C: Output and Reporting

```
Private Sub DisplayResult()
    Dim data() As String = lblResult.Tag.ToString().Split("|"c)

    Dim report As String = "STUDENT PERFORMANCE REPORT" & vbCrLf
    report &= "--------------------------" & vbCrLf
    report &= "Name: " & txtStudentName.Text & vbCrLf
    report &= "ICT: " & txtICT.Text & vbCrLf
    report &= "Visual Basic: " & txtVB.Text & vbCrLf
    report &= "Research Methods: " & txtResearch.Text & vbCrLf
    report &= "Cyber & Computer Law: " & txtCyberLaw.Text & vbCrLf
    report &= "--------------------------" & vbCrLf
    report &= "Total Marks: " & data(0) & "/400" & vbCrLf
    report &= "Average: " & Format(Val(data(1)), "0.00") & "%" & vbCrLf
    report &= "Overall Grade: " & data(2) & vbCrLf
    report &= "Status: " & data(3) & vbCrLf

    lblResult.Text = report
End Sub

Private Sub btnClear_Click(sender As Object, e As EventArgs) Handles btnClear.Click
    txtStudentName.Clear()
    txtICT.Clear()
    txtVB.Clear()
    txtResearch.Clear()
    txtCyberLaw.Clear()
    lblResult.Text = ""
End Sub

Private Sub btnSave_Click(sender As Object, e As EventArgs) Handles btnSave.Click
    Try
        Dim filePath As String = "C:\StudentResults\" & txtStudentName.Text & ".txt"
        My.Computer.FileSystem.WriteAllText(filePath, lblResult.Text, False)
        MessageBox.Show("Results saved successfully!", "Saved", MessageBoxButtons.OK, MessageBoxIcon.Information)
```